

Subextensions of Galois extensions

M/K : Galois with Galois gr G
 $\parallel$
 $\text{Gal}(M/K)$

Proposition (1): M
 $|$
 $L \leftarrow \text{sub-extension}$
 $|$
 K

Then: (1) M/L is also Galois

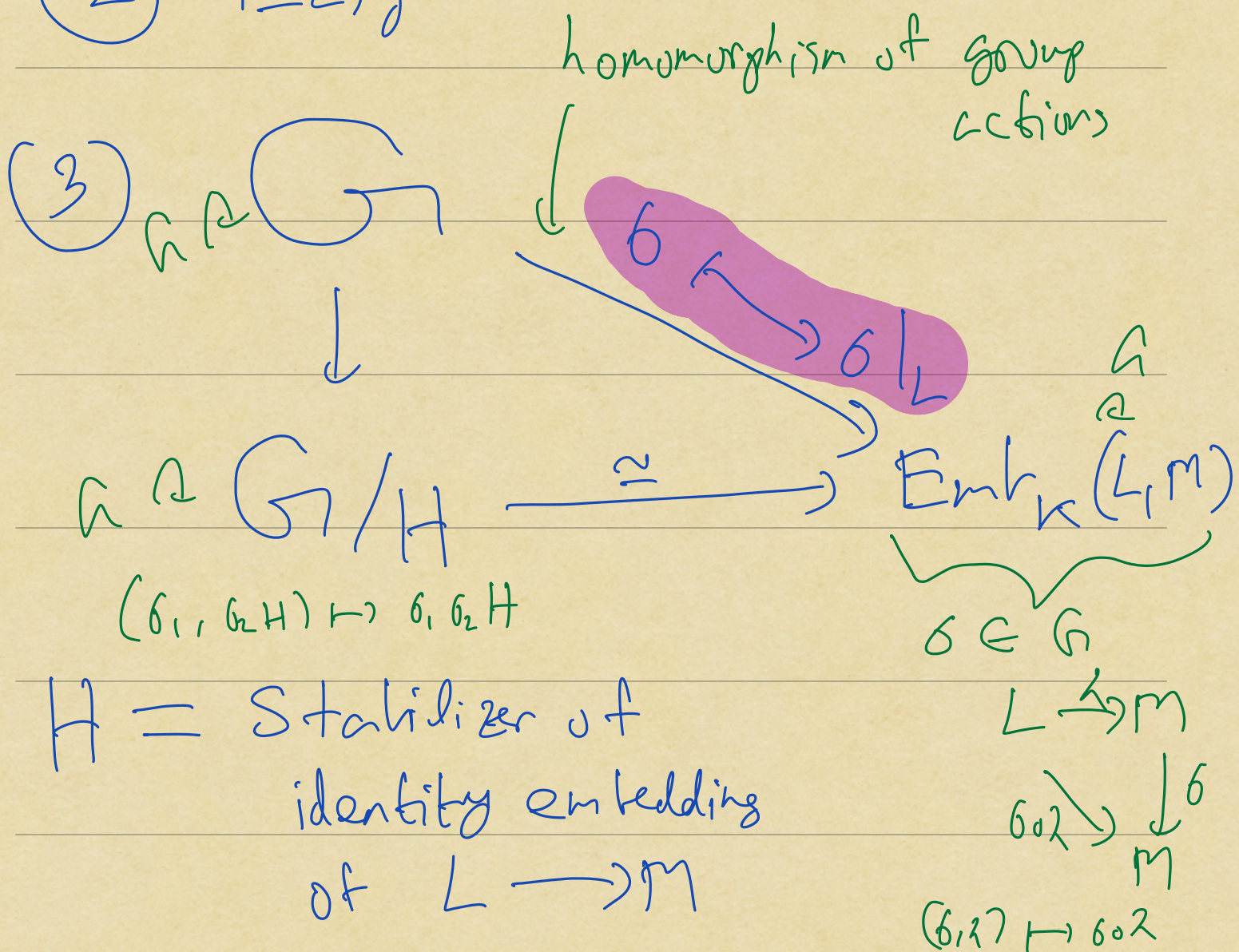
$$(2) \quad \text{Gal}(M/L) = \{ \sigma \in G : \sigma|_L = \text{id} \}$$
$$\parallel$$
$$H \leq \text{Gal}(M/K) = G$$

$$(3) \quad G/H \xrightarrow{\sim} \text{Emb}_K(L, M)$$
$$\parallel$$
$$\text{Gal}(M/K) / \text{Gal}(M/L)$$

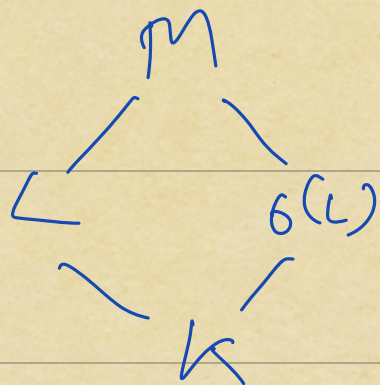
$$\sigma \longmapsto \sigma|_L$$

Pf: (1) Can use the splitting field characterization of Galois extns

(2) Easy



Proposition (2): $\sigma \in G, \sigma(L) \subseteq M$



$$\begin{aligned} (1) \quad \text{Gal}(M/\sigma(L)) &= \sigma \text{Gal}(M/L) \sigma^{-1} \\ &\leq \text{Gal}(M/K) \end{aligned}$$

(2) TFAE:

$$(a) \quad \text{Gal}(M/L) \leq \text{Gal}(M/K) \quad \text{normal}$$

$$(b) \quad L = \sigma(L), \quad \forall \sigma \in \text{Gal}(M/K)$$

$$(c) \quad L/K \text{ is Galois}$$

Pf: (1) is on HW 10 (problem 5)

(2) (b) $\Rightarrow$ (c) follows from (1)

(2) (b) $\Leftarrow$ (c)

L/K is Galois

$$\Leftarrow | \text{Emb}_K(L, L) | = [L:K]$$

$$| \text{Emb}_K^1(L, M) |$$

$$(\Leftarrow) \text{Emb}_K(L, L) = \text{Emb}_K(L, M)$$

$$(\Leftarrow) \forall \sigma \in \text{Gal}(M/K) \\ (\sigma|_L)(L) \leq L$$

$$(\Rightarrow) \forall \sigma \in \text{Gal}(M/K), \\ \text{By dimension reasons} \quad \sigma(L) = L$$

To see (a) $\Rightarrow$ (b), we will need to know more, namely the

Corollary below.

$$\text{Gal}(M/L) \text{ normal} \Rightarrow \begin{array}{c} \text{Gal}(M/\sigma(L)) \\ \parallel \\ \sigma \text{Gal}(M/L) \sigma^{-1} \\ \parallel \\ \text{Gal}(M/L) \end{array} \quad \forall \sigma \in G$$

$$\Rightarrow \sigma(L) = L, \forall \sigma \in G$$

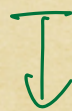
Rmk: We have constructed

$$\{ L/K : \text{subextensions of } M/K \}$$



$$\{ \text{subgroups of } \underbrace{\text{Gal}(M/K)}_G \}$$

$$L/K$$



$$\text{Gal}(M/L)$$

$$\leq \text{Gal}(M/K)$$

Defⁿ $H \leq G = \text{Gal}(M/K)$

The fixed field of H is

$$M^H = \{ \alpha \in M : \sigma(\alpha) = \alpha \quad \forall \sigma \in H \}$$

Observation: M^H/K is a subextension

pf: $K \subseteq M^H$ & M^H is closed under

Rmk (2)

L/K

$\{ \text{Subextensions of } M/K \}$

$\downarrow$

$\text{Gal}(M/L)$

$\downarrow$
 $\{ \text{Subgroups of } G = \text{Gal}(M/K) \}$

H

$\downarrow$

M^H

$\downarrow$
 $\{ \text{Subextensions of } M/K \}$

Thm (Main thm of Galois theory)

These assignments are inverses to each other. That is:

(1)

$$L = M^{\text{Gal}(M/L)}$$

$$(2) \quad H = \text{Gal}(M/M^H)$$

Corollary: The assignment

$$L|K \mapsto \text{Gal}(M/L)$$

is a bijection

$$\left\{ \begin{array}{c} \text{Subextensions} \\ L|K \\ \text{of } M|K \end{array} \right\} \xrightarrow{\cong} \left\{ \begin{array}{c} \text{Subgroups} \\ H \leq G \end{array} \right\}$$

Observation (2)

We always have inclusions

$$\bullet L \leq M^{\text{Gal}(M/L)}$$

$$\bullet H \leq \text{Gal}(M/M^H)$$

$$\{ \sigma \in G : \sigma|_{M^H} = \text{id} \}$$

Need to check that these two inclusions are bijections.

Proposition: M/K Galois

$$K = M^{G(M/K)}$$

Pf:

$$L = M^{G(M/K)} \quad |G(M/L)| = [M:L]$$

$[L:K]$

K

But $G(M/K)$ fixes L

$$\Rightarrow |G(M/L)| \geq |G(M/K)|$$

$$\Rightarrow [M:L] = [M:K]$$

$$\Rightarrow [L:k] = 1$$

$$\Rightarrow K = L \quad \checkmark$$